

1. Which of the following statements about Low density lipoprotein (LDL) is correct?

- a) LDL lipoproteins are only synthesized in the liver.
- b) LDL lipoproteins mainly transport triacylglycerols to peripheral tissues.
- c) LDL lipoproteins transfer cholesterol all tissues.
- d) LDL lipoproteins cannot be endocytosed by the liver.

2. The **key regulatory enzyme** of ketondodies synthesis is-

- a) HMG- Co A synthase
- b) HMG Co A lyase
- c) HMG Co A reductase
- d) Mevalonate kinase

3. All are conditions of ketosis **EXCEPT** one –

- a) Starvation
- b) Uncontrolled diabetes mellitus
- c) High carbohydrate diet
- d) Alcohol consumption

4. All are ketonbodies **EXCEPT** one –

- a) Acetone
- b) Acetoacetate
- c) Beta-hydroxy butyrate
- d) Citric Acid

5. The key regulatory enzyme of bile acids synthesis is

- a) Acyl coA synthetase
- b) Acetyl coA carboxylase
- c) Keto acyl synthase
- d) 7-alfa Hydroxylase

6. Gallbladder juice **does not contain** which of the following

- a) Bile salts
- b) Phospholipase A2
- c) Bilirubin
- d) Lipoprotein lipase

7. Which of the following statements about the integration of fat and carbohydrate metabolism control is correct?

- a) High insulin/glucagon ratio inhibits lipogenesis in liver.
- b) High insulin/glucagon ratio activates lipolysis in muscle
- c) Low insulin/glucagon ratio activates lipolysis in adipocytes.
- d) Insulin activates liver cell membrane glucose transporters

8. All of the following statements about β -oxidation of fatty acids IS TRUE EXCEPT?

- A) It occurs in the long fasting
- B) It is stimulated by cortisol and glucagon
- C) It occurs in the mitochondria and generates acetyl-CoA
- D) It leads directly to glucose synthesis

9. Which of the following best describes the function of lipoprotein lipase (LPL)?

- A) Synthesizes triglycerides in the liver
- B) Hydrolyzes triglycerides in chylomicrons and VLDL to free fatty acids
- C) Transports cholesterol into hepatocytes
- D) Converts cholesterol to bile acids in the liver

10. During prolonged fasting or starvation, the muscles lately switches to using which of the following energy sources?

- A) Glucose
- B) Fatty acids
- C) Ketone bodies
- D) Glycogen

11. Which of the following conditions is most likely to reduce plasma VLDL levels?

- A) High-fiber diet
- B) High-carbohydrate intake
- C) High-fat diets
- D) long fasting

12. A 4-year-old child presents with failure to thrive and chronic diarrhea. Stool analysis reveals an absence of steatorrhea despite low dietary fat intake. Further testing shows that most of the fats in the child's diet are short- and medium-chain fatty acids. Which of the following best describes how these fatty acids are absorbed?

- A) They require micelles for intestinal absorption
- B) They are absorbed directly into the portal vein
- C) They are poorly absorbed by enterocytes
- D) They are packaged into lipoproteins for transport

13. A 52-year-old woman with a BMI of 34 kg/m² is concerned about her weight and has not achieved significant results with diet and exercise. Her physician prescribes a medication that inhibits gastric and pancreatic lipases, leading to decreased fat absorption. Which of the following drugs was most likely prescribed?

- A) Ezetimibe
- B) Orlistat
- C) Statins
- D) Bile acid sequestrants

14. A healthy 25-year-old man is fasting for 24 hours as part of a religious observance. Which of the following processes would NOT be expected to occur during this fasting state?

- A) Epinephrine triggers glycogen breakdown in the liver
- B) Glucagon promotes gluconeogenesis in the liver
- C) Glucagon stimulates glycogen synthesis in the liver
- D) Low baseline insulin secretion continues from pancreatic β -cells

15. A 40-year-old marathon runner completes a 36-hour fast to prepare for a metabolic physiology study. During the fasting period, ketone body metabolism is evaluated in different tissues. Which of the following tissues is INCORRECTLY paired with its role in ketone body metabolism?

- A) Skeletal muscle — production
- B) Liver — production
- C) Cardiac muscle — utilization
- D) Kidney — utilization

16. A 58-year-old man with hyperlipidemia is started on a drug that lowers cholesterol by inhibiting HMG-CoA reductase, the rate-limiting enzyme in cholesterol biosynthesis. Which of the following drugs was most likely prescribed?

- A) Orlistat
- B) Statins
- C) Ezetimibe
- D) Choline

17. Which vitamin is required as a coenzyme for transamination?

- a) Nicotinamide adenine dinucleotide (NAD)
- b) Pyridoxal-5'-phosphate (PLP)
- c) Flavin mononucleotide (FMN)
- d) Flavin adenine dinucleotide (FAD)

18. Which amino acids are exclusively ketogenic?

- a) Alanine and cysteine
- b) Histidine and valine
- c) Leucine and lysine
- d) Tyrosine and proline

19. Maple syrup disease is a disorder due to a deficiency of pathway that degrades:

- a) Tyrosine
- b) Methionine
- c) Tryptophan
- d) Isoleucine

20. Identify the amino acid that participates in methyl group transfer:

- a) Methionine
- b) Lysine
- c) Serine
- d) Valine

21. Histamine, a chemical mediator of allergies and anaphylaxis, is synthesized from amino acid Histidine by which of the following reaction?

- a) Demethylation
- b) Hydrogenation
- c) Decarboxylation
- d) Oxygenation

22. Amino acids are released into the portal system by:

- a) Pinocytosis
- b) Active transport
- c) Facilitated diffusion
- d) Passive transport

23. For the ATP-dependent proteasome system, which of the following proteins is required to degrade proteins?

- a) Cyclin
- b) Cytochrome
- c) Ubiquitin
- d) Histone

24. Which of the following enzyme is deficient in Phenylketonuria (PKU)?

- a) Homogentisate dioxygenase
- b) p-Hydroxyphenyl- pyruvate dioxygenase
- c) Phenylalanine hydroxylase
- d) Tyrosinase

25. Which of the following enzyme is small enough to pass through the glomeruli of the kidneys?

- a) Acid Phosphatase
- b) Amylase
- c) Nucleotidase
- d) Aminotransferase

26. What enzyme in the body is measured to diagnose rhabdomyolysis?

- a) Lipase
- b) Nucleotidase
- c) Creatine kinase
- d) Glutamyltransferase

27. Which of the following are the most important cause of increased alkaline phosphatase activity in serum?

- a) liver disease, heart attack
- b) heart attack, prostate disease
- c) liver disease, bone disease
- d) bone disease, prostate disease

28. All of the following are pancreatic specific enzyme, except?

- a) Lipase
- b) Aldolase
- c) Trypsin
- d) Amylase

29. What are the main products made by the pentose phosphate pathway?

- a) ATP and NADH
- b) NADH and FADH₂

c) NADPH and ribose-5-phosphate

d) Pyruvate and acetyl-CoA

30. What happens in muscles when too much lactate builds up during hard exercise?

a) Hyperglycemia

b) Metabolic alkalosis

c) Muscle fatigue

d) Hypercalcemia

31. What can happen if the liver has impaired gluconeogenesis?

a) Persistent hyperglycemia

b) Recurrent hypoglycemia

c) Increased muscle mass

d) Increased protein synthesis

32. Which enzyme deficiency causes low ATP in red blood cells of a 5-year-old boy with tiredness and anemia?

a) Glucokinase

b) Pyruvate kinase

c) Aldolase

d) Enolase

33. The cellular location of the pyruvate dehydrogenase (PDH) complex is:

a) Cytoplasm

b) Endoplasmic reticulum

c) Mitochondrial matrix

d) Intermembrane space

34. Which molecule is the immediate precursor in glycogen synthesis?

a) Glucose-6-phosphate

b) UDP-glucose

c) Glucose-1-phosphate

d) ATP

35. Pompe disease (Type II glycogen storage disease) results from a deficiency of which enzyme?
- a) Glycogen synthase
 - b) Acid maltase**
 - c) Branching enzyme
 - d) Debranching enzyme
36. Glucose-6-phosphate dehydrogenase (G6PD) deficiency most commonly leads to:
- a) Hyperglycemia
 - b) Hemolytic anemia**
 - c) Hyperlipidemia
 - d) Ketoacidosis
37. The most common enzyme deficiency causing classic galactosemia is:
- a) Galactokinase
 - b) Galactose-1-phosphate uridylyltransferase**
 - c) UDP-glucose pyrophosphorylase
 - d) Hexokinase
38. Which cells depend solely on anaerobic glycolysis for ATP production due to the absence of mitochondria?
- a) Hepatocytes
 - b) Mature Red blood cells**
 - c) Neurons
 - d) Muscle cells
39. Where does glycogenesis mainly occur?
- a) Cytosol of liver and skeletal muscle**
 - b) Mitochondria of muscle
 - c) Nucleus of liver
 - d) Cytosol of kidney
40. Which glucose transporter is insulin-dependent and found in adipose tissue and skeletal muscle?
- a) GLUT-1
 - b) GLUT-2**

- c) GLUT-3
- d) GLUT-4

41. Which of the following is necessary to convert heme into bilirubin in the body?

- a) Glucuronic acid and Aminolevulinic acid
- b) Carbon monoxide and glutamine
- c) NADPH and oxygen
- d) glucuronic acid and NADH

42. Inhibition of which of the following enzymes is used in the treatment of gout?

- a) Thymidylate synthetase
- b) Adenosine deaminase
- c) Xanthine oxidase
- d) Ribonucleotide reductase

43. All the following compounds are involved in the biosynthesis of both purine and pyrimidine nucleotides except:

- a) Glutamine
- b) aspartate
- c) Glycine
- d) carbon dioxide

44. Which step of the following is the regulatory step in the Heme synthesis done?

- a) Delta aminolevulinic formation
- b) Conversion of delta-aminolevulinic acid to porphobilinogen
- c) Cyclization of tetrapyrrole
- d) Decarboxylation of uroporphyrinogen

45. Porphyria cutanea tarda is caused by a defect in which of the following enzymes?

- a) ALA dehydratase
- b) Porphobilinogen synthase

- c) Porphyrinogen oxidase
- d) Uroporphyrinogen decarboxylase

46.5- Fluorouracil inhibits which of the following enzymes?

- a) Ribonucleotide reductase
- b) Dihydrofolate reductase
- c) Thymidine synthase
- d) Xanthine oxidase

47. Uric acid levels increase in all of the following, except:

- a) Fanconi syndrome
- b) Lash-Nihan syndrome
- c) cancer patients
- d) gout

48. IMP dehydrogenase in the synthesis of purine nucleotides is inhibited by which of the following?

- a) Mycophenolic acid
- b) Methotrexate
- c) Allopurinol
- d) 6- Mercaptopurine

49. Which of the complexes of the respiratory chain does not act as a proton pump?

- a) Succinate Q Reductase
- b) NADH dehydrogenase
- c) Cytochrome C oxidase
- d) Cyt C reductase

50. The ATP/ADP transporter in the mitochondrial inner membrane is inhibited by

- a) Oligomycin
- b) CN⁻
- c) Atracetylloside
- d) Barbiturates

51. Which of the transporters of the respiratory chain has the lowest oxidizing power?

- a) FMN
- b) Coenzyme Q
- c) Cytb
- d) Cytc

52. In the respiratory chain, cytochrome a is located in complex.....

- a) I
- b) II
- c) III
- d) IV